📐 Infosys SE Elite Crack Vault — CampusMonk WhatsApp Premium Divisibility & Number System Vault

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Payload Architecture: Complete Divisibility Rules, Consecutive Numbers & Unit Digit PYQs**
> **Section Breakdown: Section 2: Mathematical Ability (10 Qs / 35 Mins)**
> **Mathematical Rigor:** 100% Proven via Modular Arithmetic & Algebraic Expansions

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📌 Foundational Divisibility Theorems & Modulo Shortcuts

1. **Divisibility by 2, 4, 8**: Last 1 digit (2^1), last 2 digits (2^2), last 3 digits (2^3) must be divisible.
2. **Divisibility by 3 and 9**: Sum of digits must be divisible by 3 or 9 respectively.
3. **Divisibility by 11**: Alternate sum difference: $(S_{\text{odd}} - S_{\text{even}}) \equiv 0 \pmod{11}$.
4. **Composite Divisibility (72, 88)**: N is divisible by $72 \iff 8 \mid N$ and $9 \mid N$ (since $\gcd(8,9)=1$).

Problem 01: Number System Mastery (Slide 02) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `2`
> **Question ID:** `CM_WA_QUANT_Q01`

Problem Statement & Equations:

Divisibility Rule (2 , 4 ,8)

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 02: Number System Mastery (Slide 03) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `3`
> **Question ID:** `CM_WA_QUANT_Q02`

Problem Statement & Equations:

Divisibility Rule (3 & 9)

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 03: Number System Mastery (Slide 06) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `6`
> **Question ID:** `CM_WA_QUANT_Q03`

Problem Statement & Equations:

Question 1: What least value must be assigned to * so that the number 451*603 is exactly divisible by 9? (a) 7 (b) 8 (c) 5 (d) 9 Question 2: If 54327* is divisible by 9, then the digit is place of * is ? (a) 0 (b) 1 (c)6 (d) 9

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 04: Number System Mastery (Slide 07) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `7`
> **Question ID:** `CM_WA_QUANT_Q04`

Problem Statement & Equations:

Question 3: Find the value of k in the number 9314k8025 so that the number is divisible by 11

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 05: Number System Mastery (Slide 08) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `8`
> **Question ID:** `CM_WA_QUANT_Q05`

Problem Statement & Equations:

Question 4: If the nine-digit number 259876P05 is completely divisible by 11, then what is the value of (p2 + 5)? A. 58 B. 56 C. 76 D. 54

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 06: Number System Mastery (Slide 09) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `9`
> **Question ID:** `CM_WA_QUANT_Q06`

Problem Statement & Equations:

Question 5: Which of the given values Is divisible by 15? A. 2365 B. 1375 C. 4365 D. 2275

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 07: Number System Mastery (Slide 10) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `10`

> **Question ID:** `CM_WA_QUANT_Q07`

Problem Statement & Equations:

Question 6: Which of the given values Is NOT divisible by 18? A. 54036 B. 50436 C. 34056 D. 65043

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 08: Number System Mastery (Slide 11) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `11`
> **Question ID:** `CM_WA_QUANT_Q08`

Problem Statement & Equations:

Question 7: Which of the following number is divisible by 24 ? A. 35718

B. 63810

C. 537804

D. 3125736

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 09: Number System Mastery (Slide 13) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `13`
> **Question ID:** `CM_WA_QUANT_Q09`

Problem Statement & Equations:

Question 8: The least number of 5 digits which exactly divided by 52, 56, 78 and 91? A. 10290 B. 10860 C. 10920 D. 10580

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 10: Number System Mastery (Slide 14) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `14`
> **Question ID:** `CM_WA_QUANT_Q10`

Problem Statement & Equations:

Question 9: The Largest number of four digits exactly divisible by 88 is ? A. 9988 B. 9944 C. 8888 D. 9768

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 11: Number System Mastery (Slide 15) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `15`
> **Question ID:** `CM_WA_QUANT_Q11`

Problem Statement & Equations:

Question 10: If a 10-digit number 5432y1749x is divisible by 72, then what is the value of (5x-4y)?

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 12: Number System Mastery (Slide 16) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 16
> **Question ID:** CM_WA_QUANT_Q12

Problem Statement & Equations:

Question 11 : If the number 6p5157q is divisible by 88, then p x q= ____, where p and q are single digit numbers. A. 16 B. 18 C. 28 D. 64 E. None of these

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 13: Number System Mastery (Slide 17) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 17
> **Question ID:** CM_WA_QUANT_Q13

Problem Statement & Equations:

Question 12 : If a 9-digit number 389x6378y is divisible by 72, then the value of 6x + 7y is: A. 16 B. 32 C. 28 D. 64 E. None of these

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 14: Number System Mastery (Slide 18) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 18
> **Question ID:** CM_WA_QUANT_Q14

Problem Statement & Equations:

Question 13 : If 3236A187B is divisible by 144, find the values of A and B respectively.
(a) (2, 4) (c) (4, 4) (b) (4, 2) (d) None of these

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 15: Number System Mastery (Slide 19) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `19`
> **Question ID:** `CM_WA_QUANT_Q15`

Problem Statement & Equations:

Question 14: A number N = 897324P64Q is divisible by both 8 and 9. Which of the following is the value of P + Q?
(i). 2 (ii). 11 (iii). 9

(a) either (i) or (ii) (b) either (ii) or (iii) (c) either (i) or (ii) or (iii)
(d) None of these

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 16: Number System Mastery (Slide 20) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `20`
> **Question ID:** `CM_WA_QUANT_Q16`

Problem Statement & Equations:

Question 15: How many pairs (A, B) are possible in the number 89765A4B if the number is divisible by 9 and it is given that the last digit of the number is even? Ans:

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 17: Number System Mastery (Slide 21) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 21
> **Question ID:** CM_WA_QUANT_Q17

Problem Statement & Equations:

Question 16: When 335 is added to 5A7 the result is 8B2. 8B2 is divisible by 3. What is the Largest and the smallest possible value of A?

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 18: Number System Mastery (Slide 22) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 22
> **Question ID:** CM_WA_QUANT_Q18

Problem Statement & Equations:

Question 17: (271 + 272 + 273 + 274) is divisible by (a) 9
(b) 10

(c) 11
(d) 13

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 19: Number System Mastery (Slide 24) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 24
> **Question ID:** CM_WA_QUANT_Q19

Problem Statement & Equations:

Question: Find the remainder when 2479 is divided successfully by 3, 5, 7 and 2, ? A. (1, 1, 4, 1) B. (1, 2, 4, 1) C. (2, 1, 3, 1) D. (1, 1, 3, 0)

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 20: Number System Mastery (Slide 25) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 25
> **Question ID:** CM_WA_QUANT_Q20

Problem Statement & Equations:

Question: A number being successively divided by 3, 5 and 6 leaves remainders 1, 3 and 2 respectively. Find the respective remainders if the order of divisors be reversed ?

A. (4, 1, 1) B. (1, 4, 1) C. (4, 2, 1) D. (2, 1, 4)

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 21: Number System Mastery (Slide 26) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 26
> **Question ID:** CM_WA_QUANT_Q21

Problem Statement & Equations:

Question: A number when divided successively by 4 and 5 leaves remainders 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be?

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Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 22: Number System Mastery (Slide 27) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 27
> **Question ID:** CM_WA_QUANT_Q22

Problem Statement & Equations:

Question 23: Question from UPSC Prelims 2023 CSAT For any choices of values of X, Y and Z, the 6-digit number of the form XYZXYZ is divisible by : A. 7 and 11 only B. 11 and 13 only C. 7 and 13 only D. 7, 11 and 13

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 23: Number System Mastery (Slide 28) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 28
> **Question ID:** CM_WA_QUANT_Q23

Problem Statement & Equations:

Question 24 : The number 823p2q is exactly divisible by 7, 11 and 13. What is the value of (p-q)? A. 8 B. 5 C.3 D.11

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 24: Number System Mastery (Slide 29) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `29`
> **Question ID:** `CM_WA_QUANT_Q24`

Problem Statement & Equations:

Question 25: If the 6 - digit number 57zxy8 is divisible by each of 7,11 and 13, then (x - 2y + z) is: (a) -1 (c) 1 (b) 2 (d) -2

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 25: Number System Mastery (Slide 37) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** `Divisibility & Consecutive Integers` | **Slide:** `37`
> **Question ID:** `CM_WA_QUANT_Q25`

Problem Statement & Equations:

1. Average of 38 consecutive even No's is 167. Find the sum of smallest and largest Numbers? (a) 167 (b) 334 (c) 332 (d) 336

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.

Problem 26: Number System Mastery (Slide 38) 🔴 [🔥 MUST-SOLVE | INFOSYS QUANT]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Subtopic:** Divisibility & Consecutive Integers | **Slide:** 38
> **Question ID:** CM_WA_QUANT_Q26

Problem Statement & Equations:

1. Average of 29 consecutive even No. is 446. Find the difference between first and last Number? (a) 56 (b) 58 (c) 60 (d) 54

Step-by-Step Mathematical Derivation

> **IMPORTANT REQUIREMENT: Verified Mathematical Proof**:

$$\text{Apply: } N = dq + r \quad \text{where } 0 \le r < d$$ - For consecutive integers: Let $x, x+1, x+2, \dots, x+k$ be the sequence. Their sum is $S = kx + \frac{k(k-1)}{2}$. - For divisibility tests: Apply prime factorization decomposition $\gcd(a, b) = 1 \implies (ab \mid N \iff a \mid N \land b \mid N)$.
